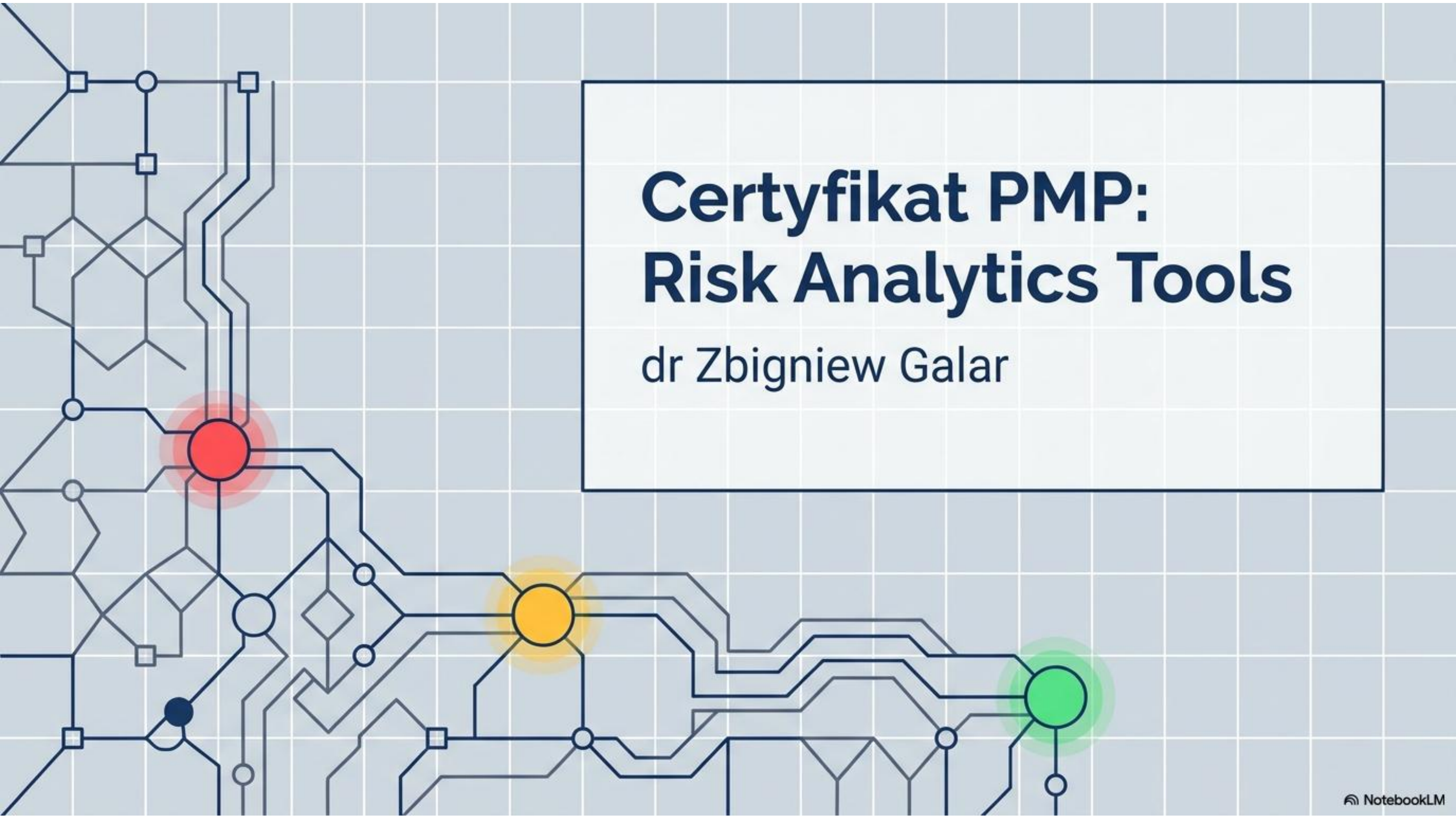


The background of the slide is a light blue grid with a complex, abstract circuit diagram overlaid on the left side. The diagram consists of various geometric shapes like squares, circles, and diamonds connected by lines, with some nodes highlighted in red, yellow, and green.

Certyfikat PMP: Risk Analytics Tools

dr Zbigniew Galar

Two Engines of Risk Analysis



Qualitative Assessment

Speed & Cost

Fast, relatively easy, cost-effective

Nature of Data

Subjective (Expert Judgment)

Application

Applied to ALL identified risks

Primary Output

Prioritized rank (High/Med/Low)

High

Medium

Low



Quantitative Assessment

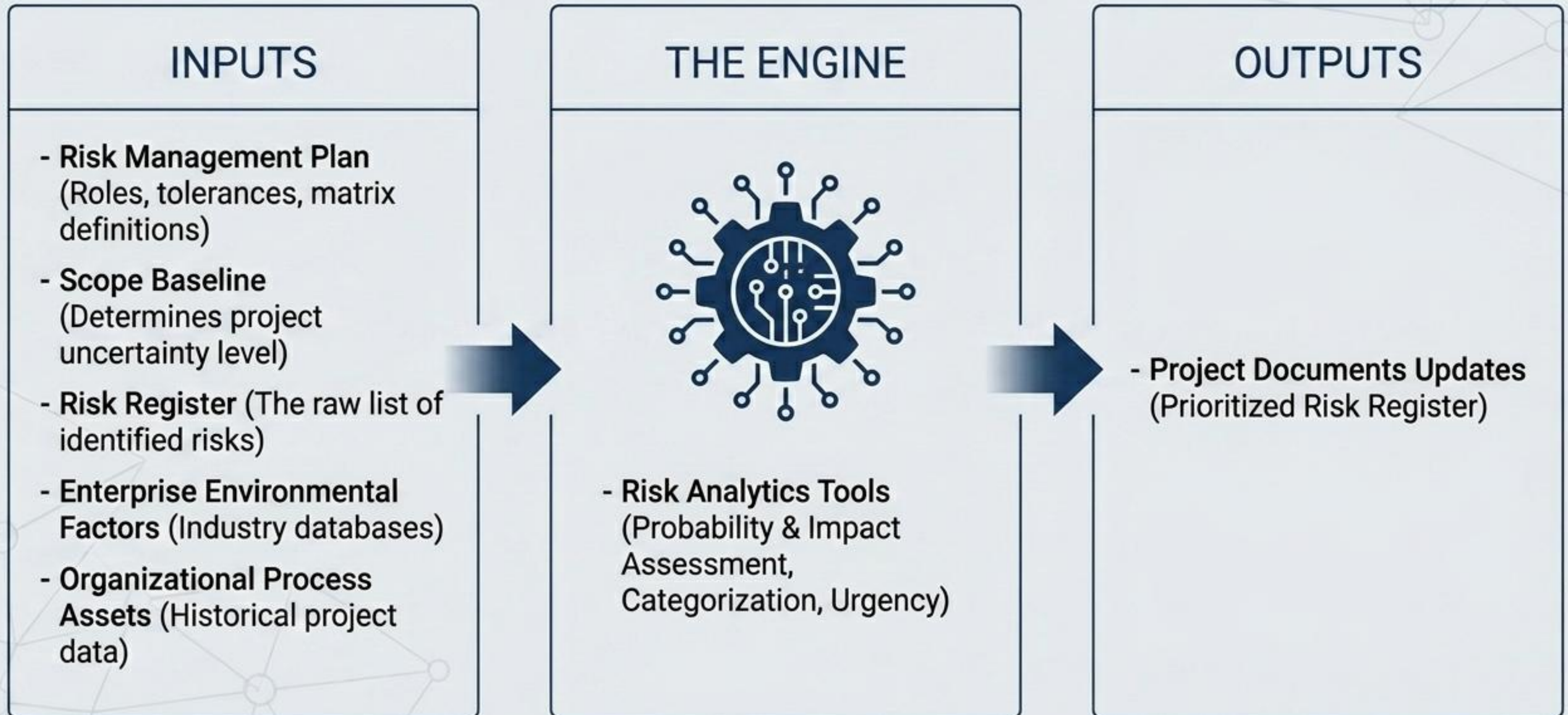
Time-consuming, complex, higher cost

Objective (Numeric, Statistical)

Applied ONLY to high-priority risks

Aggregate project impact (Schedule/Cost)

The Architecture of Qualitative Analysis



Defining the Axes of Risk Measurement

Probability Card



- **Definition:** The likelihood an event will occur.
- **Scale:** 0.0 (No probability) to 1.0 (100% certainty).
- **Rule:** Probability of occurrence + Probability of non-occurrence always equals 1.0.
- **Method:** Determined via Expert Judgment (past experiences and historical data).

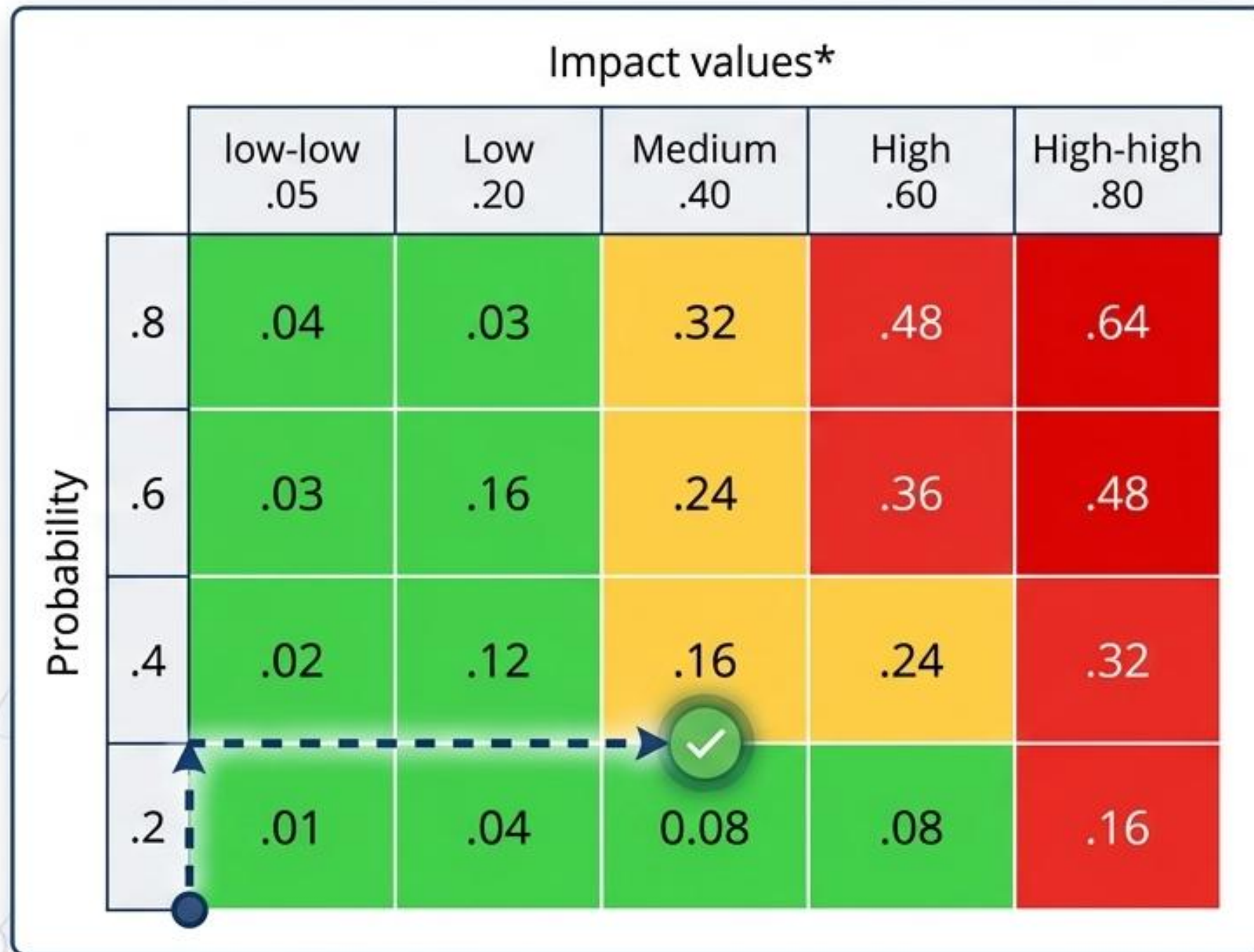
Impact Card

Objectives	Low-low (0.05)	Low (0.20)	Medium (0.40)	High (0.60)	High-high (0.80)
Cost	No significant impact	Less than 6% increase	7-12% increase	13-18% increase	More than 18% increase
Time	No significant impact	Less than 6% increase	7-12% increase	13-18% increase	More than 18% increase
Quality	No significant impact	Few components impacted	Significant impact requiring customer approval to proceed	Unacceptable quality	Product not usable

- **Definition:** The amount of pain or gain posed to project constraints (Cost, Time, Quality).
- **Ordinal Scales:** Relative ranking (High, Medium, Low).
- **Cardinal Scales:** Numeric increments (Linear or Nonlinear).
- **Example Thresholds (Cost):**
 - Low (0.20): < 6% increase
 - Medium (0.40): 7-12% increase
 - High (0.60): 13-18% increase

The Probability & Impact Heatmap

Intersecting the axes yields an overall risk score and priority classification.



Probability Value: 0.2

Impact Score: 0.40 (Medium)

Overall Risk Score: 0.08
(Low / Green Condition)

- Organization sets matrix rules and thresholds prior to the process.
- Guides the Plan Risk Responses phase—only high/red risks receive formal quantitative analysis.

Case Study: Screen Scrapers, Inc.

Step 1: The Context



- **Project:** Converting legacy mainframe green screens to Windows-compatible browser screens.
- **Challenge:** High uncertainty, entry-level users, inconsistent navigation.

Step 2: The Method



- **Delphi Technique:** An anonymous panel of internal company experts and external industry specialists.
- **Process:** Email summaries sent, probability and impact assigned via predefined scales.

Step 3: The Raw Output



- Vendor viability
- Vendor responsiveness
- Software compatibility
- Hardware compatibility
- Mainframe connection
- Staff training

Scoring the Screen Scrapers Matrix

Impact Scale Baseline

High-high = 0.8 | High = 0.6 | Medium = 0.4 | Low = 0.2 | Low-low = 0.05

Custom Probability & Impact Matrix

Probability	Impact scores*	.20	.40	.60	.80
.8	.04	.16	.32	.48	.64
.6	.03	.12	.24	.36	.48
.4	.02	.08	.16	.24	.32
.2	.01	.04	.08	.12	.16

Final Risk Mapping

- Hardware compatibility: $0.6 \text{ Prob} \times 0.8 \text{ Impact} = 0.48$ **HIGH**
- Mainframe connection: $0.2 \text{ Prob} \times 0.8 \text{ Impact} = 0.16$ **MEDIUM**
- Vendor viability: $0.6 \text{ Prob} \times 0.6 \text{ Impact} = 0.36$ **HIGH**
- Vendor responsiveness: $0.4 \text{ Prob} \times 0.4 \text{ Impact} = 0.16$ **MEDIUM**
- Software compatibility: $0.4 \text{ Prob} \times 0.4 \text{ Impact} = 0.16$ **MEDIUM**
- Staff training: $0.2 \text{ Prob} \times 0.05 \text{ Impact} = 0.01$ **LOW**

Refining Qualitative Assessments

Risk Data Quality Assessment



Must validate reliability, integrity, and accuracy of gathered data. Low-quality data renders prioritization useless.

Risk Categorizations



Groups risks by project phase or Work Breakdown Structure (WBS) to identify which project elements are most exposed.

Risk Urgency Assessment



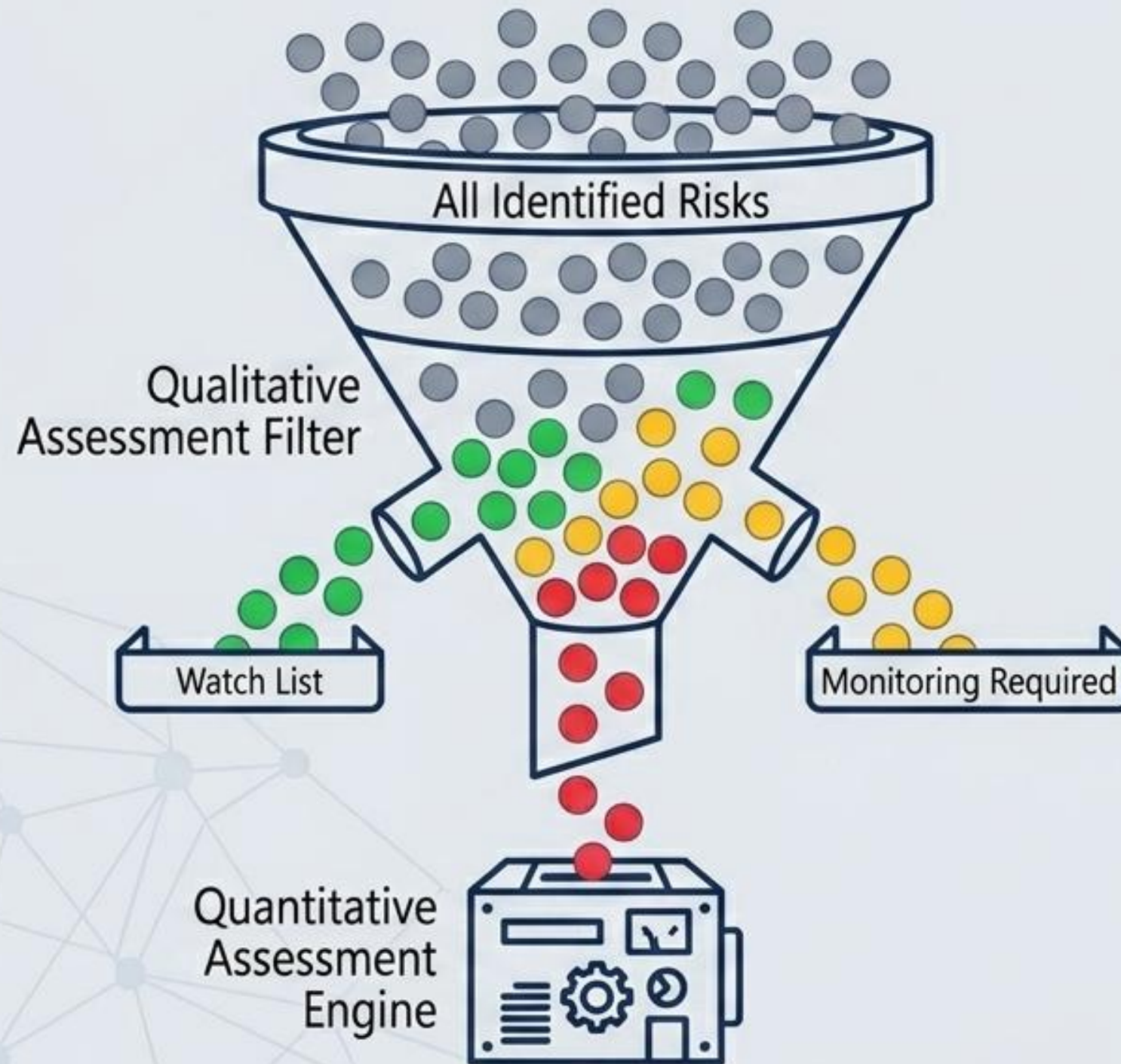
Evaluates risk triggers and response times to flag risks requiring near-term action regardless of overall score.

Expert Judgment & Bias Check



Evaluations rely heavily on subjective expertise (interviews/workshops). The PM must actively identify and correct expert bias.

The Risk Funnel: Filtering for Impact



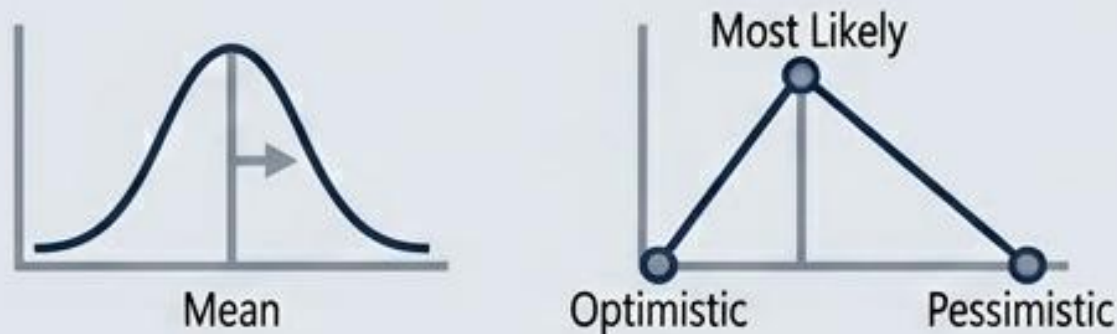
- 100% of identified risks undergo Qualitative Analysis.
- Low-priority (Green) risks enter the Watch List.
- Medium-priority (Yellow) risks require monitoring.
- High-priority (Red) risks drop into the Quantitative phase to determine aggregate schedule, cost, and scope impact.

Quantifying Aggregate Risk Exposure

Goal: Assign numeric probabilities to identify realistic cost/schedule targets.

Probability Distributions Glossary

Continuous Distributions (Time/Cost Models):



- Triangular: Uses 3-point estimates (Pessimistic, Most Likely, Optimistic).
- Normal & Lognormal: Uses mean and standard deviations to quantify risk.
- Beta & Uniform: Utilized for variables with specific graphical distribution requirements.

Discrete Distributions (Scenario Models):

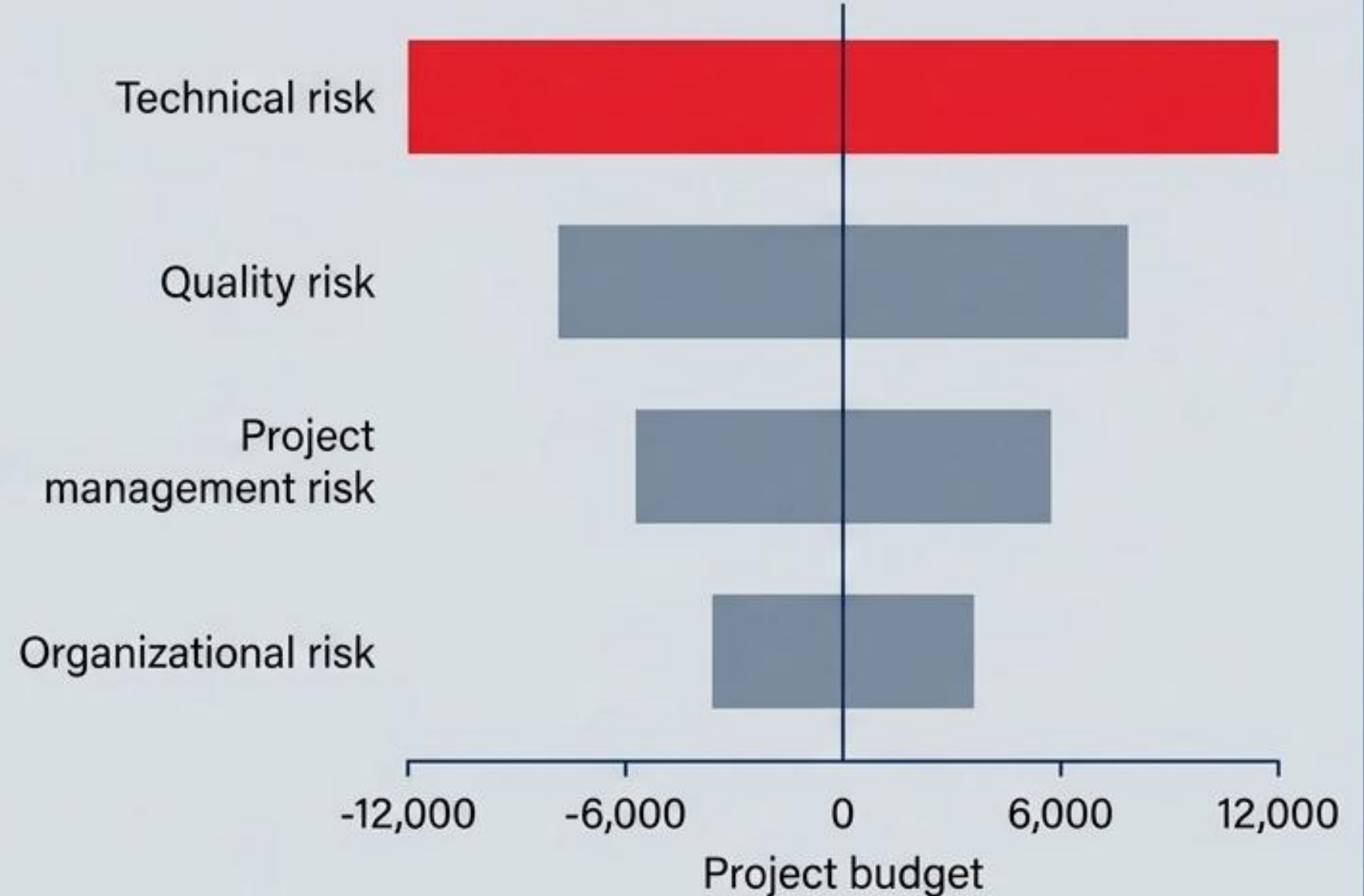


- Represents specific, distinct possible scenarios.
- Used primarily in Decision Trees, prototype outcomes, and pass/fail tests.

Sensitivity Analysis: The Tornado Diagram

Isolating variables with the greatest potential impact

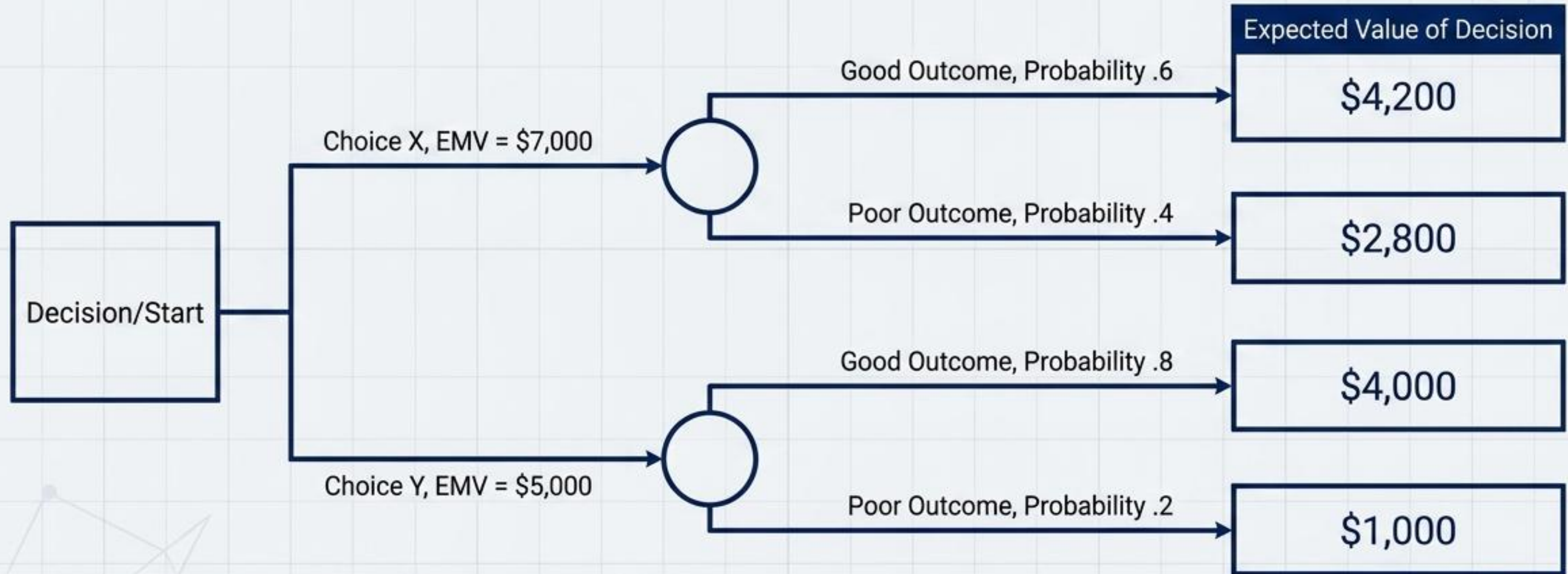
- Mechanism: Examines all uncertain elements at their baseline values to determine which individual risk has the largest effect on the project.
- Structure: Horizontal bars represent a sensitivity variable (low to high value limits).
- Insight: Variables at the top pose the largest threat to time, cost, or quality objectives, dictating where detailed response plans are required.



Decision Tree Analysis & EMV

Calculating the average, anticipated future impact of a decision under uncertainty.

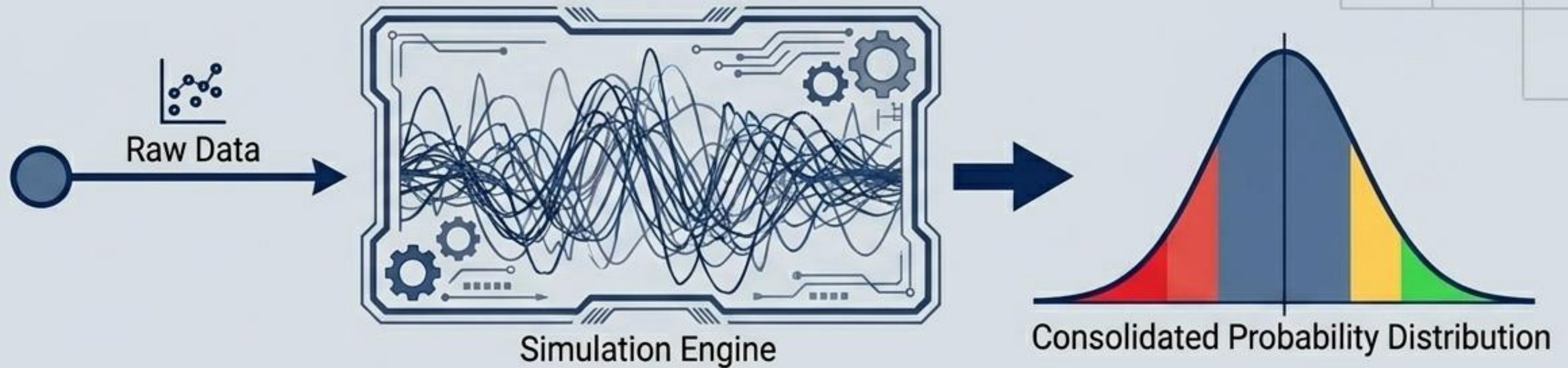
Decision Formula:
 $\text{Probability of Risk} \times \text{Impact} = \text{Expected Monetary Value (EMV)}$



- Decision Trees diagram the interrelated sequence of choices.
- Positive results indicate project opportunities.
- Negative results indicate project threats.
- The optimal path is the choice with the greatest net expected value.

Modeling & Simulation: Monte Carlo Analysis

Predicting schedule and cost risks from the perspective of the whole project.



Methodology & Inputs

- The Method: Computes the project model thousands of times using varying inputs (cost estimates or activity durations).
- The Variables: Cost risks utilize the Work Breakdown Structure (WBS); Schedule risks utilize the schedule network diagram.

Outputs & Benefits

- The Output: Generates a complete probability distribution for total project cost or duration.
- Advantage: Highly recommended for predicting cost/schedule risk as it is more powerful and less prone to misuse than EMV.

The Synthesis: The Risk Register Dashboard

All analytical tools ultimately serve a single, living document.

